WASP-52b

R-0447

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-52_171026` · created 2026-07-19T15:41:10+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458052.6538 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.013
Transit depth (Rp/Rs)^2	3.23 +/- 0.47 [%]
Orbital Inclination (inc)	84.53 +/- 0.57
Airmass coefficient 1 (a1)	1.102 +/- 0.011
Airmass coefficient 2 (a2)	-0.085 +/- 0.02
Scatter in the residuals of the lightcurve fit is	0.97 %
Best Comparison Star	None
Optimal Aperture	4.51
Optimal Annulus	8.21
Transit Duration (day)	0.0719 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.18 ± 0.013 vs 0.1646 (deviation 1.18σ)
Duration fit vs published	0.0719 d vs 0.0758 d (deviation 0.89σ)
Mid-time O–C	6.6 min
Depth SNR	13.8
Residual scatter	0.97 %

Light curve

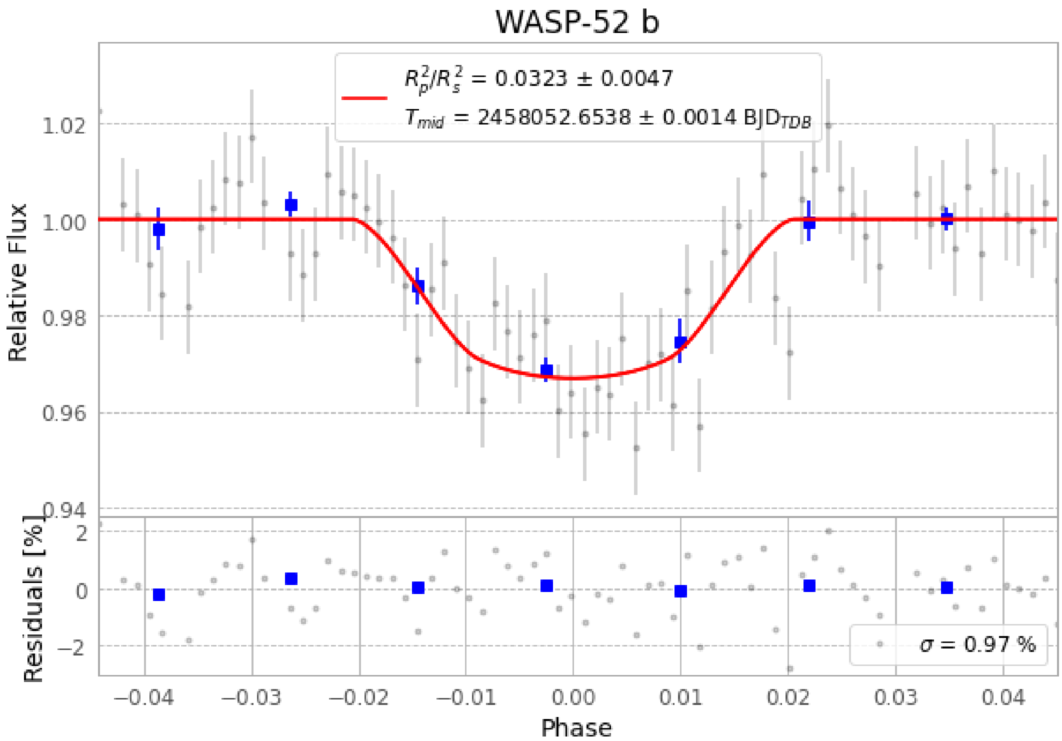

Figure 1 — FinalLightCurve_WASP-52 b_2017-10-25.png (SHA-256 0f6fa297af901f10…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [327, 189], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 cffd027a1ed460f6…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

74 FITS frames (49 MB), `WASP-52171026014212.FITS` → `WASP-52171026052712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-171026.log (fb94b14f9c5b…), FinalLightCurve_WASP-52 b_2017-10-25.png (0f6fa297af90…), FinalLightCurve_WASP-52 b_2017-10-25.csv (b14ce9fc02f3…)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-19 15:41Z.